

AAKRIT SHARMA

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EDUCATION

Thapar Institute of Engineering & Technology

2023 – 2027

B.E. Computer Engineering

CGPA: 8

SUMMARY

Computer Engineering undergraduate focused on **AI Engineering, LLM systems, agentic workflows, and scalable inference infrastructure**. Experienced in building **retrieval systems, memory architectures, ranking engines, and production-oriented AI applications**. Strong foundation in **Data Structures & Algorithms, Backend Engineering, NLP, Distributed Systems, and Model Deployment** using Python, PyTorch, FastAPI, Docker.

EXPERIENCE

Outstyl

Aug 2025 – Sep 2026

Machine Learning Intern

- Developed transformer-based recommendation systems using **BERT4Rec** and **SASRec** for predictive user behavior modeling and ranking optimization.
- Built scalable **PyTorch** training pipelines for sequential recommendation and representation learning workflows.
- Improved recommendation quality through temporal embeddings, transformer ranking architectures, and evaluation pipelines.

PROJECTS

DualSpeechLM Optimization Pipeline

GitHub

LLMs, Multimodal AI, LoRA, Quantization, PyTorch

- Designed adaptive token-routing and context-compression modules for efficient multimodal speech-language model inference.
- Integrated **Phi-3.5-mini-Instruct** using **4-bit NF4 quantization** and LoRA fine-tuning, significantly reducing memory requirements while preserving performance.
- Built GRU-based dialogue-state tracking, semantic importance scoring, and prosody-aware compression pipelines for long-context conversations.
- Developed auxiliary objectives for dialogue-act classification, semantic alignment, and emotion-aware response generation.

Persistent Context Memory Store

GitHub

MCP, Agentic AI, Context Engineering, Retrieval Systems

- Built a local-first memory system enabling AI agents and IDE assistants to share persistent context through the **Model Context Protocol (MCP)**.
- Developed automated memory-generation pipelines that convert conversations and project artifacts into structured Markdown-based knowledge repositories.
- Implemented semantic retrieval, memory ranking, and context selection mechanisms for long-horizon agent workflows.
- Maintained a unified global memory layer across development environments, reducing context fragmentation between AI tools.

AI Candidate Ranking Engine

GitHub

Embeddings, LLMs, FastAPI, Docker, Ranking Systems

- Designed a large-scale candidate evaluation pipeline processing **100,000+ resumes** using embeddings, ranking signals, and cross-attention architectures.
- Built precomputation, indexing, and retrieval workflows enabling low-latency candidate search and ranking.
- Integrated local LLM-based reasoning modules to generate explainable hiring recommendations and ranking justifications.
- Containerized the inference stack using Docker for reproducible deployment and scalable serving.

ACHIEVEMENTS

- Top 100 / 250,000+ Teams** – Adobe India Hackathon for building an AI-powered document intelligence platform.
- NVIDIA Certified** – Accelerated Data Science using CUDA and RAPIDS.
- Winner, **CTRL+SPACE Hackathon** among **2,500+ participants**.
- Solved **300+ DSA problems** across algorithms, graphs, dynamic programming, and advanced data structures.

SKILLS

Languages	Python, C++, JavaScript, SQL, C, Solidity, MATLAB
AI Engineering	LLMs, RAG, Agentic AI, Context Engineering, Prompt Engineering, Retrieval Systems
ML / DL	PyTorch, TensorFlow, Transformers, NLP, Reinforcement Learning, Scikit-learn
Infrastructure	FastAPI, Docker, Linux, Git, REST APIs, MCP
Data Systems	PostgreSQL, SQLite, Redis, Embeddings, Vector Search, Ranking Systems
LLM Optimization	HuggingFace, PEFT, LoRA, Quantization, Model Evaluation, Inference Optimization
Core CS	Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks